

CraftNext

MAJOR PROJECT REPORT

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Bachelors of Technology in Computer Science & Engineering

Submitted To:



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DECLARATION

We the undersigned solemnly declare that the project report “CraftNext” is based on my own work carried out during the course of our study under the supervision of Prof. Gayathri Naidu, Professor , CSE .

We assert the statements made and conclusions drawn are the outcomes of my own work. I further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree / diploma / certificate in this university or any other University of India or abroad.
3. We have followed the guidelines provided by the university in writing the report.

Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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In this semester, we have completed our project on “CraftNext”. During this time, all the group members collaboratively worked on the project and learnt about the industry standards that how projects are being developed in IT Companies. We also understood the importance of teamwork while creating a project and got to learn the new technologies on which we are going to work in near future.

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We perceive this as an opportunity and a big milestone in our career development. We will strive to use gained skills and knowledge in our best possible way and we will work to improve them.

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ABSTRACT

In traditional artisan economies and cultural handicraft sectors, small-scale craftspeople and rural artisans face severe market fragmentation and exploitative multi-tier intermediary networks. Handcrafted arts such as Lippan mirror work, terracotta pottery, handloom textiles, and brass crafts represent centuries of cultural heritage; however, traditional retail supply chains divert up to 60–70% of product margins to brokers, physical exhibitions, and middlemen. Furthermore, mainstream commercial e-commerce platforms impose prohibitive listing fees, strict onboarding hurdles, and algorithmic bias toward industrial mass production, leaving grassroots artisans marginalized and economically underserved.

The CraftNext – Digital Marketplace for Handcrafted Products is an end-to-end web-based commerce platform engineered specifically to eliminate exploitative intermediaries and provide direct artisan-to-consumer digital connectivity. The platform delivers an authentic, culturally immersive shopping interface where conscious buyers can discover verified artisans, explore rich craft provenance stories, browse categorized indigenous crafts, and purchase directly from original creators without intermediary markup.

The system allows consumers to register and authenticate securely via email OTP, search and filter crafts across multiple criteria, consult a conversational shopping assistant powered by Google Gemini AI, manage wishlists, and execute idempotent checkout transactions. Upon order confirmation, the platform dynamically generates and streams itemized PDF invoices using PDFKit, providing comprehensive billing transparency and real-time delivery lifecycle tracking.

In addition, CraftNext provides a comprehensive Artisan Portal empowering rural and boutique creators with autonomous storefront management. Artisans can easily register, list unique handcrafted products, upload multi-angle photography with automated image compression, configure category classifications, track real-time stock levels, and monitor customer orders through an intuitive dashboard designed for maximum accessibility.

The CraftNext platform is architected using modern web technologies adhering to a decoupled three-tier model: a responsive client-side Single-Page Application (HTML5, Vanilla CSS design tokens, and Progressive Web App service worker caching for offline resilience), an asynchronous Node.js and Express.js RESTful backend, and a cloud-hosted MongoDB database utilizing Mongoose ODM. Critical transactional mechanisms such as MongoDB replica-set ACID sessions ensure atomic stock decrements, while Razorpay payment gateway integration provides reliable, secure online transactions.

The system aims to enhance economic empowerment for grassroots artisans, protect intangible cultural heritage, eliminate commercial counterfeit confusion, and offer a transparent, socially conscious purchasing journey. By bridging traditional artisan heritage with contemporary web standards, CraftNext demonstrates how modern digital technology can foster sustainable socio-economic growth and preserve cultural artistry for future generations.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CSE	Computer Science and Engineering
CDN	Content Delivery Network
CRUD	Create, Read, Update and Delete
DB	Database
DFD	Data Flow Diagram
ER	Entity Relationship
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express.js, React.js / HTML5, Node.js
MVC	Model View Controller
ODM	Object Data Modeling
OTP	One-Time Password
PWA	Progressive Web Application
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Artisan commerce and handcrafted production are foundational components of India's cultural heritage and rural economic livelihood. Across diverse states and tribal clusters, over seven million skilled artisans produce exquisite handloom textiles, Lippan mirror art, terracotta pottery, brassware, and wood carvings. Whenever traditional creators seek to market their work beyond local village boundaries, an organized commercial channel becomes indispensable to connect makers with contemporary urban and international buyers.

In a traditional handicraft system, an artisan relies upon localized middlemen, traveling brokers, and regional wholesale aggregators. Although this approach is historically familiar, it becomes deeply exploitative and inefficient. Multiple tiers of intermediaries extract up to 60–70% of final retail margins, leaving the original creator impoverished while inflating prices for consumers. This project specifically identifies rural artisan clusters, craft cooperatives, independent boutique studios, and cultural consumers as environments where conventional distribution creates acute economic hardship.

The problem becomes even more severe in the modern digital era. Mainstream commercial e-commerce platforms impose prohibitive commissions (15% to 30%), strict barcode compliance, rigid inventory penalties, and search ranking algorithms tailored strictly for mass-manufactured factory goods. As a result, authentic handmade crafts are either excluded from digital markets or undercut by industrial plastic and ceramic counterfeits.

Another major limitation of traditional supply chains is the complete absence of cultural storytelling and transparency. Consumers purchasing traditional handicrafts rarely know who created the piece, what indigenous techniques were employed, or whether the price paid genuinely supports the maker's household. Digital technologies provide an unparalleled opportunity to bridge this gap through virtual storefronts and direct-to-maker commerce.

CraftNext – Digital Marketplace for Handcrafted Products is developed with this objective. It provides an integrated platform for artisan discovery, cultural heritage storytelling, and direct e-commerce transactions. The platform allows consumers to register, browse authentic craft categories, view artisan origin profiles, select unique products, execute idempotent checkout transactions, and track live order progress with automated invoice delivery.

The project follows a modern full-stack architecture. HTML5, Vanilla CSS design tokens, and Progressive Web App (PWA) caching are used for frontend development. Node.js and Express.js are used for backend services, and a MongoDB database hosted through MongoDB Atlas is used for structured transactional data storage. Email OTP handles

authentication, Razorpay provides secure online payment processing, Google Gemini AI provides conversational shopping recommendations, and PDFKit handles dynamic invoice generation and delivery.

The primary purpose of CraftNext is not merely to create an online storefront. It establishes an integrated artisan empowerment ecosystem where rural creators gain sovereign digital storefronts, receive fair remuneration, and share the intangible cultural heritage behind their crafts with a global audience.

1.2 Problem Definition

Traditional handicraft distribution commonly relies on opaque multi-tier intermediary chains. Creators produce items in isolated villages, wait months for visiting traders, accept deeply discounted payments, and forfeit all control over retail pricing. This creates several problems.

1.2.1 Severe Middleman Exploitation

Artisans typically receive only 20% to 30% of the final consumer retail price, while brokers, regional wholesalers, and boutique retail showrooms absorb the remaining 70% as commercial margin.

1.2.2 Lack of Direct Digital Storefront Access

Rural artisans lack access to lightweight, user-friendly digital tools to independently showcase their portfolios, manage orders, and communicate directly with consumers.

1.2.3 Counterfeit Proliferation and Authenticity Crisis

Consumers face widespread uncertainty regarding product genuineness due to the influx of cheap, machine-made replicas falsely branded as authentic handloom or tribal art.

1.2.4 Inefficient Inventory and Concurrency Hazards

Handcrafted items are inherently produced in limited batches. Concurrent purchases without atomic database locks lead to overselling and customer dissatisfaction.

1.2.5 High Commission Barriers on Mainstream Portals

General marketplaces charge prohibitive listing fees, monthly subscriptions, and 15–30% commissions that small independent creators cannot afford.

1.2.6 Absence of Cultural Storytelling

Commercial catalogs present products as generic stock items without highlighting the artisan's biography, community heritage, and intricate crafting techniques.

1.2.7 Inefficient Logistics and Manual Invoicing

Manual order fulfillment and lack of automated billing generate compliance headaches and impede scaling for growing craft cooperatives.

Problem Statement

To design and develop a web-based Digital Marketplace for Handcrafted Products that allows consumers to discover authentic crafts, explore artisan heritage stories, execute atomic and idempotent purchases, and track order fulfillment in real time, while providing rural and boutique artisans with an accessible, low-friction portal for storefront management, inventory synchronization, and direct revenue generation.

1.3 Motivation

The motivation behind CraftNext can be divided into three major categories.

1.3.1 User Motivation

Modern conscious consumers actively seek authentic, ethical, and eco-friendly handcrafted products. A digital craft platform allows users to bypass commercial counterfeits, discover makers directly, verify material origins, consult intelligent AI shopping assistants, and enjoy a seamless e-commerce experience from any mobile or desktop browser.

1.3.2 Artisan Motivation

Grassroots creators require economic sovereignty. CraftNext empowers artisans with direct customer reach, 100% transparent pricing, automated inventory tracking, instant payment settlement, and centralized order processing, eliminating dependency on exploitative middlemen.

1.3.3 Technical Motivation

From a software-development perspective, CraftNext provides an opportunity to implement multiple modern technologies in a single real-world application, including decoupled SPA design, Node.js REST APIs, MongoDB ACID transactions, Razorpay, and Gemini AI.

1.4 Objectives

The main objective of CraftNext is to develop a digital platform that simplifies craft commerce and empowers artisans. The specific objectives are:

- To develop a web-based digital craft marketplace.
- To eliminate intermediary markups and ensure direct artisan payouts.
- To provide digital storefront management for rural creators.
- To enable authentic cultural storytelling and maker provenance display.
- To integrate conversational AI craft recommendations via Google Gemini.
- To engineer atomic stock decrement using MongoDB ACID transactions.
- To implement idempotent order checkout preventing duplicate charges.
- To allow users to filter crafts according to categories.

- To provide automated, publication-ready PDF invoice generation via PDFKit.
- To allow artisans to manage incoming customer requests.
- To provide order/service status management.
- To improve communication between users and artisans.
- To provide secure email OTP authentication and session management.
- To create a responsive and user-friendly interface.
- To develop a scalable software architecture.
- To demonstrate the practical application of modern full-stack technologies.

1.5 Scope of the Project

CraftNext is primarily designed for environments where traditional artisan crafts require direct market reach. Potential application areas include:

1. Rural Artisan Clusters

Empowering regional craft communities with direct global market access.

2. Conscious Urban & Global Consumers

Offering verified authentic, ethically sourced handmade decor and gifts.

3. Artisan Cooperatives and SHGs

Enabling craft collectives to manage joint inventory and aggregate shipping.

4. Independent Boutique Studios

Providing emerging handicraft designers with a zero-commission storefront.

5. Cultural Tourism and Heritage Centers

Preserving traditional folk art forms through digital documentation.

1.5.1 Existing System

The existing traditional system generally follows the process: Artisan Production → Local Middleman → Regional Wholesaler → Retail Showroom → Consumer Purchase.

For example, consider a terracotta artisan in rural Gujarat: the artisan crafts clay utensils over several weeks, sells them at nominal rates to a visiting trader, who marks up the items by 400% before selling to urban consumers.

Advantages:

- Simple and easy to understand for rural creators.
- No specialized software or digital devices required.

Disadvantages:

- Severe margin extraction of up to 60–70% by middlemen.
- Delayed, unpredictable payments to artisans.
- Complete loss of product provenance and artisan identity.
- Rampant proliferation of counterfeit machine replicas.
- Artisans have zero visibility into final consumer demand.
- Vulnerability to unsold inventory and storage losses.
- High entry and commission barriers on commercial portals.

1.5.2 Proposed System

The proposed CraftNext system transforms the traditional process into an integrated digital workflow:

User Registration/Login → Artisan Onboarding → Product Listing → Browse/Search → Product Selection → View Craft Story → Add to Cart → Idempotent Checkout → Razorpay Transaction → Atomic Stock Decrement → PDF Invoice Stream → Order Notification → Artisan Fulfillment → Real-Time Timeline Tracking → Service Completion.

Advantages of Proposed System:

- 100% direct remuneration to makers with zero intermediary fee.
- Direct consumer-to-artisan transparency and cultural storytelling.
- Atomic inventory management eliminating concurrent overselling.
- Conversational AI craft recommendations via Google Gemini.
- Dynamic, publication-ready PDF invoice generation via PDFKit.
- Responsive Progressive Web App accessible on desktop and mobile.

1.6 Hardware and Software Requirements**1.6.1 Hardware Requirements****Minimum Hardware**

Component	Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Storage	At least 10 GB free
Internet	Required
Display	Standard monitor / mobile screen
Input	Keyboard and Mouse / Touchscreen

Recommended Hardware

Component	Requirement
Processor	Intel Core i5 or higher
RAM	8 GB or higher
Storage	SSD (Solid State Drive)
Internet	High-speed broadband / 4G
Display	Full HD recommended

1.6.2 Software Requirements

Software/Technology	Purpose
Windows / Linux / macOS	Operating System
HTML5 & Vanilla CSS	Frontend markup and UI styling
JavaScript (ES6+)	Client-side reactivity and async operations
Progressive Web App (PWA)	Offline asset caching and installability
Node.js	Backend runtime environment
Express.js	Backend REST API framework
MongoDB	Document-oriented database
MongoDB Atlas	Cloud database hosting
Mongoose ODM	Schema modeling and ACID transactions
Razorpay API	Online payment processing and verification
Google Gemini AI	Conversational natural language recommendations
PDFKit	Dynamic binary invoice streaming engine
Visual Studio Code	Integrated development environment
Git / GitHub	Version control and code repository
Chrome / Edge / Firefox	Cross-browser testing and validation

CHAPTER 2

LITERATURE SURVEY

2.1 Study of Existing Systems

Handicraft retail and artisan market systems have evolved across traditional physical bazaars and modern digital commerce. While conventional systems rely on physical exhibitions, digital marketplaces have introduced global cataloging, electronic payment gateways, and automated logistics. Retail, artisan cooperatives, and e-commerce platforms are commonly analyzed to benchmark digital craft solutions.

2.1.1 Traditional Physical Artisan Bazaars and Haats

In a physical craft haat, artisans travel to urban exhibitions (e.g., Dilli Haat) and display wares directly. The workflow follows Arrival → Stall Setup → In-person Negotiation → Cash Payment → Physical Handover. The system is simple but creates significant travel, accommodation, and seasonal limitations.

2.1.2 General Commercial E-Commerce Platforms

Multi-vendor platforms such as Amazon and Flipkart allow consumers to discover thousands of retail brands. However, their listing standards, rigid packaging guidelines, high commissions (15–30%), and algorithms biased toward mass-manufactured items heavily disadvantage small artisan producers [1].

2.1.3 Dedicated Global Craft Platforms

Global platforms such as Etsy provide dedicated handmade portals. Advantages include international exposure and high buyer trust. Limitations include per-listing charges, foreign currency friction, high courier fees, and severe competition from industrialized factory replicas.

2.1.4 Social Media Commerce

Boutique creators frequently sell through Instagram and WhatsApp. While visually engaging, social commerce lacks automated payment verification, centralized inventory control, formal invoice generation, and customer dispute safeguards.

2.1.5 Government and Tribal Portals

State-sponsored platforms (e.g., TRIFED, Tribes India) assist tribal artisans by procuring stock. While protective, these portals frequently suffer from outdated user interfaces, sluggish catalog refreshes, and bureaucratic payment cycles, with the main objective of supporting rural livelihoods.

availability and improving artisan market reach. CraftNext applies modern decoupled web principles to streamline and modernize these artisan sales channels.

2.1.6 Artisan Cooperatives and Self-Help Groups

Cooperatives organize local makers to collectively market handloom and craft goods, demonstrating how pooling digital resources can expand commercial reach.

2.2 Advantages, Disadvantages and Limitations of Existing Systems

Existing System	Advantages	Disadvantages
Physical Haats	Direct interaction, authentic touch	Heavy travel cost, seasonal, localized
General E-Commerce	High traffic, integrated logistics	High commission (15–30%), mass-goods bias
Global Portals (Etsy)	International reach, high buyer intent	Per-listing fees, forex barriers, drop-shipping
Social Commerce	Visual, zero setup fee	Manual orders, no payment guarantee, unorganized
Government Portals	Fair base price, state backing	Outdated UI, slow payouts, rigid cataloging
CraftNext Platform	Zero listing fees, direct payout, AI search	Requires smartphone / internet connectivity

The literature reviewed concludes that digital solutions can improve artisan margins and preserve cultural heritage, while also identifying a gap between systems focused only on mass retail and systems preserving cultural storytelling.

2.3 Technology Background

2.3.1 HTML5 & Vanilla CSS Design Tokens

HTML5 and Vanilla CSS design tokens provide lightweight, high-performance UI rendering without heavy external JavaScript bundle bloat [2]. Custom variables establish a cohesive design system across all pages.

- Zero compile-step build overhead.
- Consistent glassmorphic design token management.
- Ultra-fast first contentful paint (FCP).
- Native cross-browser compatibility.
- Large development ecosystem.

Role in CraftNext: HTML5 and Vanilla CSS are used to build the responsive user and artisan interfaces.

2.3.2 Progressive Web Applications (PWA)

PWA standards combine the broad accessibility of the web with native app functionality using Service Workers.

- Cache-first asset delivery for instant startup.
- Home screen installation without app store friction.
- Offline browsing resilience for previously viewed crafts.
- Low data consumption on mobile devices.

Role in CraftNext: PWA ensures offline resilience and mobile-friendly usability for rural artisans.

2.3.3 Node.js

Node.js provides an asynchronous, event-driven JavaScript runtime environment on the server side [3], handling concurrent I/O operations efficiently.

Role in CraftNext: Node.js executes server-side operations and API handling.

2.3.4 Express.js

Express.js is a minimalist web framework used for routing, middleware pipelines, API development, and request handling [3].

Role in CraftNext: Express acts as the backend framework serving 28 RESTful endpoints.

2.3.5 MongoDB & Mongoose ODM

MongoDB is a document database storing JSON-like BSON documents [4]. Mongoose ODM provides schema validation and replica-set multi-document ACID transactions for atomic stock decrement.

2.3.6 Email OTP Authentication

Email OTP authentication utilizes cryptographically secure 6-digit codes and signed JSON Web Tokens (JWT) [6] to handle registration, login, and access control.

2.3.7 Razorpay Payment Gateway

Razorpay provides comprehensive digital payment infrastructure supporting UPI, Net Banking, credit/debit cards, and wallets. In CraftNext, payments are validated using cryptographic HMAC-SHA256 signature verification before order confirmation, ensuring complete transactional security.

2.3.8 Google Gemini AI & PDFKit

Google Gemini AI powers conversational shopping assistance [7], answering material inquiries and suggesting craft gifts. PDFKit handles dynamic binary invoice generation, streaming itemized tax invoices directly to buyers upon purchase.

2.4 Comparative Analysis

Feature	Traditional Store	General E-Com	Etsy	CraftNext
Digital Access	No	Yes	Yes	Yes
Direct Artisan Payout	No (Middlemen)	No (15–30% fee)	Partial	Yes (100% direct)
Zero Listing Fees	No	No	No	Yes
Cultural Storytelling	Limited	No	Limited	Yes (Rich stories)
Conversational AI	No	Generic bot	No	Yes (Gemini AI)
Atomic Concurrency	Manual	Yes	Yes	Yes (ACID tx)
PWA Offline Support	No	Partial	No	Yes (Full PWA)
Dynamic PDF Invoices	Manual receipt	Automated	Automated	Yes (Streamed)
Live Order Tracking	No	Yes	Yes	Yes (Real-time)

2.5 Proposed System: Advantages, Disadvantages and Limitations

Advantages:

- Direct-to-Maker Economics – 100% of sale proceeds reach artisans.
- Transparent Provenance – Cultural context and artisan origin stories accompany every piece.
- Atomic Stock Integrity – Prevents double-booking during concurrent shopping bursts.
- AI-Powered Discovery – Gemini AI provides personalized craft guidance.
- Instant Tax Invoicing – PDFKit streams publication-ready invoices on checkout.
- Responsive Design – The system can be accessed through desktop and mobile browsers.

Disadvantages:

- Internet connectivity is required.
- Users need access to a digital device.
- Artisans require basic technical knowledge.
- The system requires ongoing cloud maintenance.
- Cloud services may introduce operational dependencies.

Limitations:

- The current project is primarily web-based.
- Domestic shipping is prioritized in the current implementation.
- AI-based 3D room placement is future scope.
- Large-scale deployment may require additional infrastructure.
- Notification mechanisms can be further expanded.

CHAPTER 3

METHODOLOGY

3.1 Existing Methodology

The traditional methodology is mainly manual and consists of the following steps:

1. Craft Production – The artisan creates handcrafted wares in home workshops.
2. Middleman Aggregation – A visiting broker inspects and buys wares at discounted rates.
3. Wholesaler Transit – Goods are transported to regional commercial centers.
4. Retail Exhibition – Showrooms markup products and display them to urban consumers.
5. Delayed Settlement – Artisans receive deferred payments months later.
6. Customer Purchase – The consumer buys goods without knowing maker provenance.

Major Problems:

- Severe margin loss and delayed cash flow for rural creators.
- Zero direct communication between artisan and consumer.
- No digital inventory tracking or sales records.
- Risk of unsold inventory and craft spoilage.
- Manual management and lack of data collection.

3.2 Proposed Methodology

The proposed CraftNext methodology follows a digital process:

1. User Registration/Login – The user accesses the platform and authenticates using email OTP.
2. Artisan Discovery – The user views verified artisans and their studios.
3. Search and Filtering – The user filters crafts based on categories or price range.
4. Product Selection – The user opens the selected handcrafted product.
5. Craft Story Exploration – The user reads the maker's story, materials, and origin.
6. Cart Storage – The user adds items to the persistent shopping cart.
7. Idempotent Checkout – The system validates stock and creates a unique idempotency key.
8. Payment Execution – The user completes payment via Razorpay or selects COD.
9. Atomic Stock Decrement – The system commits a MongoDB ACID transaction session.
10. Dynamic Invoicing – PDFKit renders and streams an itemized tax invoice.
11. Order Notification – The artisan receives the order alert on the seller portal.
12. Real-Time Timeline Update – The artisan updates status (Shipped, Delivered).
13. Completion – The craft is delivered and the user submits an artisan review.

3.3 System Development Process

The project development process consists of the following stages:

1. Requirement Analysis – Identify user and artisan requirements.
2. System Design – Design application architecture, database, and interface.
3. Implementation – Develop frontend, backend, and database functionality.
4. Integration – Connect frontend, backend, authentication, database, and payment services.
5. Testing – Test each module individually and collectively.
6. Deployment – Prepare the application for web-based access.

3.4 Project Modules

Module	Purpose
Authentication	User access and identity
User / Buyer	Customer interaction
Artisan Discovery	Find verified artisans
Queue & Stock Management	Manage concurrent inventory
Order Management	Manage orders and invoices
Artisan	Artisan storefront operations
Real-Time Communication	Live updates and timeline
Image Management	Artisan product images

3.4.1 Authentication Module

Responsible for registration, login, session management, user verification, and access control.

3.4.2 User Module

Responsible for browsing crafts, searching, filtering, selecting items, managing cart, and tracking orders.

3.4.3 Artisan Module

Responsible for artisan profile management, product listing, inventory management, and order status updates.

3.4.4 Queue & Stock Management Module

This is the central concurrency module of CraftNext, responsible for atomic stock decrements, idempotency, and preventing overselling.

3.4.5 Order Management Module

Responsible for order creation, order storage, artisan order viewing, status updates, invoice generation, and customer tracking.

3.5 System Architecture

The proposed architecture consists of several layers:

- **Presentation Layer — HTML5 + Vanilla CSS + PWA** – Handles UI, user interaction, forms, dashboards, craft cards, and cart interface.
- **Application Layer — Node.js + Express.js** – Handles API requests, business logic, validation, stock locks, and order processing.
- **Data Layer — MongoDB + MongoDB Atlas** – Stores user data, artisan data, catalog data, and order data.
- **External Services — Razorpay, Gemini AI, PDFKit** – Handles payment gateway, conversational craft recommendations, and invoice generation.

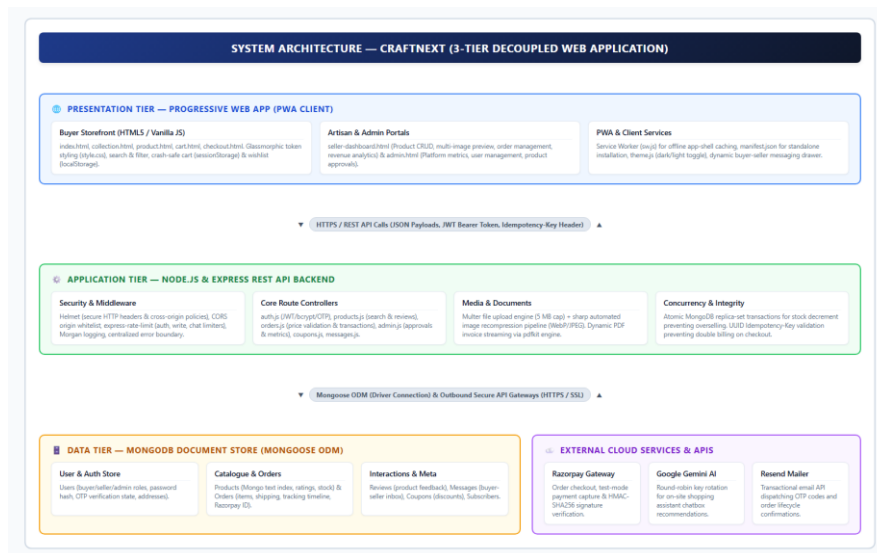


Fig. 4.6 – System Architecture / Data Flow Overview

3.6 Database Methodology

The project uses a structured document approach. The major collections/entities include:

- **User** – stores user-related information.
- **Artisan** – stores artisan profile and studio information.
- **Category** – stores craft/service categories.
- **Product** – stores handcrafted products offered by artisans.
- **Cart / Queue** – represents an active checkout reservation.
- **Order** – stores customer order information.
- **Order Item** – stores individual items associated with an order.

3.7 Real-Time Communication Methodology

Real-time communication is important because product stock and order status change frequently. The process can be represented as:

Artisan changes order status → Backend processes update → Real-time event published → Timeline updates → User receives update → User interface refreshes delivery timeline.

This reduces the need for manual page refreshing and keeps both users and artisans synchronised with the latest inventory and order state [7].

CHAPTER 4

DIAGRAMS AND IMPLEMENTATION

4.1 Data Flow Diagram

4.1.1 Context-Level DFD

External entities: User and Artisan. Central system: CraftNext Platform.

User → CraftNext: Login information, craft search, item selection, checkout, order info.

CraftNext → User: Artisan information, stock status, order status, invoices, notifications.

Artisan → CraftNext: Artisan information, product information, stock updates, fulfillment status.

CraftNext → Artisan: Customer orders, sales notifications, revenue metrics.

4.1.2 Level 0 DFD

The system is decomposed into five major processes:

- Process 1 – Authentication: User ↔ Authentication ↔ User Database
- Process 2 – Artisan Management: User ↔ Artisan Management ↔ Artisan Database
- Process 3 – Stock & Concurrency: User ↔ Stock Management ↔ Product Database
- Process 4 – Order Management: User ↔ Order Management ↔ Order Database
- Process 5 – Notification / Status Updates: Artisan → Status Update → Real-Time Service → User

4.1.3 Level 1 DFD – Order & Inventory Flow

The order process is decomposed into the following steps: receive checkout request, validate artisan, check stock availability, generate idempotency key, execute transaction, store order data, stream PDF invoice, publish update, and display delivery information.



Fig. 4.1–4.3 – Data Flow Diagram of the CraftNext System

4.2 Use Case Diagram

The Use Case Diagram represents the interaction between actors and the system. Users and artisans are the primary actors.

User Use Cases:

- Register
- Login
- Browse Artisans
- Search Craft
- Filter Craft
- View Artisan
- Select Product
- Add to Cart
- Checkout Order
- Track Order
- View Invoices
- Consult AI Assistant
- Receive Updates
- Logout

Artisan Use Cases:

- Login
- Manage Artisan Profile
- Manage Products
- View Customer Requests
- Update Stock Levels
- Process Orders
- Update Order Status
- Complete Service

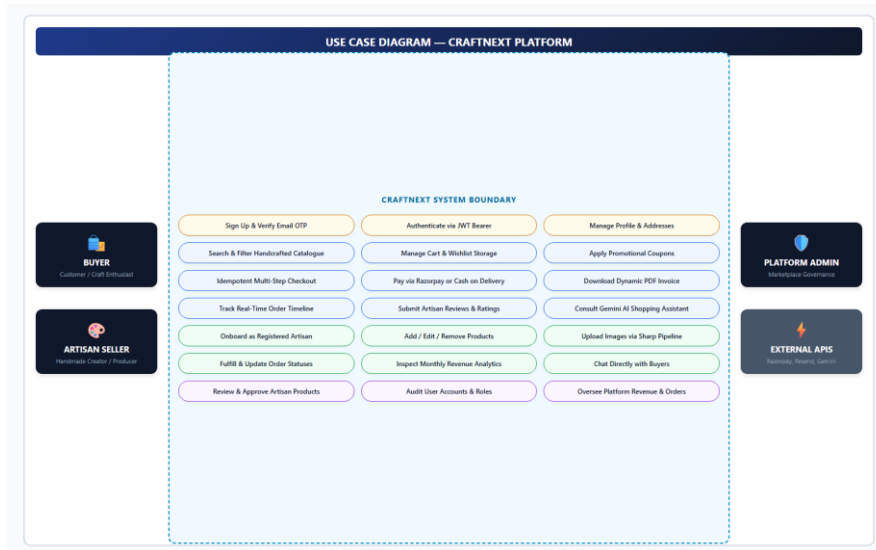


Fig. 4.4 – Use Case Diagram

4.3 E-R Diagram

The ER diagram represents the database structure and relationships between users, orders, and products [4]. Recommended entities and their key attributes are described below.

Entity	Key Attributes
USER	user_id, name, email, authentication_id, created_at
ARTISAN	artisan_id, name, description, category_id, rating, availability, image_url
CATEGORY	category_id, category_name
PRODUCT	product_id, artisan_id, name, description, price, stock, availability
CART_RESERVATION	reservation_id, product_id, user_id, status, expires_at
ORDER	order_id, user_id, artisan_id, total_amount, status, created_at
ORDER_ITEM	order_item_id, order_id, product_id, quantity, price

Relationships:

- User → places → Order
- User → manages → Cart
- Artisan → manages → Products

- Artisan $\rightarrow$ receives $\rightarrow$ Order
- Artisan $\rightarrow$ offers $\rightarrow$ Product
- Product $\rightarrow$ belongs to $\rightarrow$ Category
- Order $\rightarrow$ contains $\rightarrow$ Order Item



Fig. 4.5 – E-R Diagram

4.4 System Flowchart

Start → Open CraftNext → Register/Login → Authentication Successful? (No → Display Error → Login Again; Yes ↓) → Display Dashboard → Browse Artisans → Search/Filter → Select Artisan → View Products → Select Product → Add to Cart → Proceed to Checkout → Process Payment → Atomic Stock Decrement → Display Order Confirmation → Stream PDF Invoice → Artisan Processes Order → Update Status → Send Real-Time Update → User Receives Update → Service Completed → End.

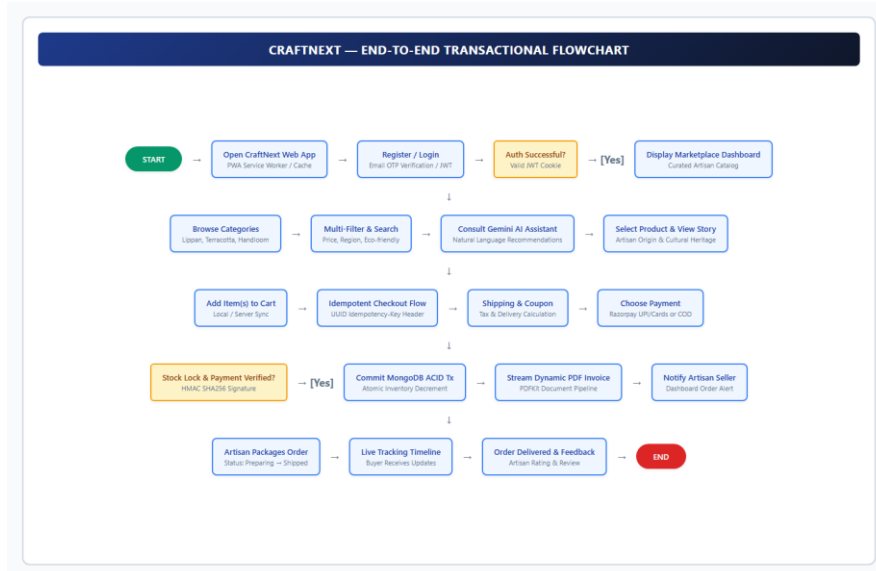


Fig. 4.7 – System Flowchart (represented via the system data-flow overview)

4.5 Module Description

4.5.1 Login and Registration Module

The login and registration interface is the entry point to the application. Users provide the required information and authenticate themselves. Main functions include registration, login, input validation, authentication, session management, and access control. The interface provides secure authentication, input validation, and responsive design [6].

4.5.2 User Dashboard

The dashboard acts as the main control center, displaying available artisans, categories, product collections, ongoing order information, notifications, and user options. It is designed to allow users to view artisans and purchase authentic crafts.

4.5.3 Artisan Discovery Module

Users can browse available artisans. An artisan card typically contains the artisan name, image, category, rating, availability, description, and an action button, making artisan discovery easier than physically visiting different workshops.

4.5.4 Cart & Checkout Interface

The checkout interface is the most important part of the project. After adding products to cart, the user executes payment, receives an order token, tracks delivery progress, and receives automated PDF invoices.

4.5.5 Artisan Dashboard

The artisan dashboard provides tools for managing customer requests, including viewing incoming orders, updating product stock, updating service status, processing customer requests, and monitoring revenue progress. It is a dedicated interface for managing incoming orders and product listings.

4.5.6 Order Management

Orders are displayed to the artisan according to status: Pending → In Progress → Completed.

4.6 Pseudocode of Modules

4.6.1 User Login

```
START
Display Login Page
INPUT email
INPUT password
Send authentication request
IF authentication successful THEN
    Create authenticated session
    Redirect user to dashboard
ELSE
    Display authentication error
END IF
STOP
```

4.6.2 Artisan Search

```
START
INPUT search keyword
Retrieve available artisans
FOR each artisan
    Compare artisan information with search keyword
    IF match found THEN
        Display artisan
    END IF
END FOR
IF no artisan found THEN
    Display "No artisans found"
END IF
STOP
```

4.6.3 Artisan Filtering

```
START
INPUT selected category
Retrieve artisans
FOR each artisan
    IF artisan category matches selected category THEN
        Display artisan
    END IF
END FOR
STOP
```

4.6.4 Join Cart & Validate Stock

```
START
User selects artisan product
CHECK product availability
IF product is unavailable THEN
    Display "Product unavailable"
    STOP
END IF
Retrieve current stock
Generate temporary reservation
Create order entry
Calculate order total
Store order entry
Publish stock update
Display confirmation and order summary
STOP
```

4.6.5 Stock & Order Update

```
START
Artisan selects next order
Find corresponding order entry
Update status
Move order to next position
Update database
Publish real-time order event
Notify connected users
STOP
```

4.6.6 Create Order

```
START
User selects product/service
Validate product availability
IF product unavailable THEN
    Display unavailable message
    STOP
END IF
Create order
Store order information
Set status = Pending
Send order to artisan
STOP
```

4.6.7 Update Order Status

```
START
Artisan selects order
IF order is pending THEN
  Change status to In Progress
ELSE IF order is in progress THEN
  Change status to Completed
END IF
Save updated status
Publish real-time update
Display updated status to user
STOP
```

4.7 Screenshots of Modules

4.7.1 Login and Registration Interface

This interface allows users to create accounts and authenticate themselves, providing clear input fields and appropriate validation.

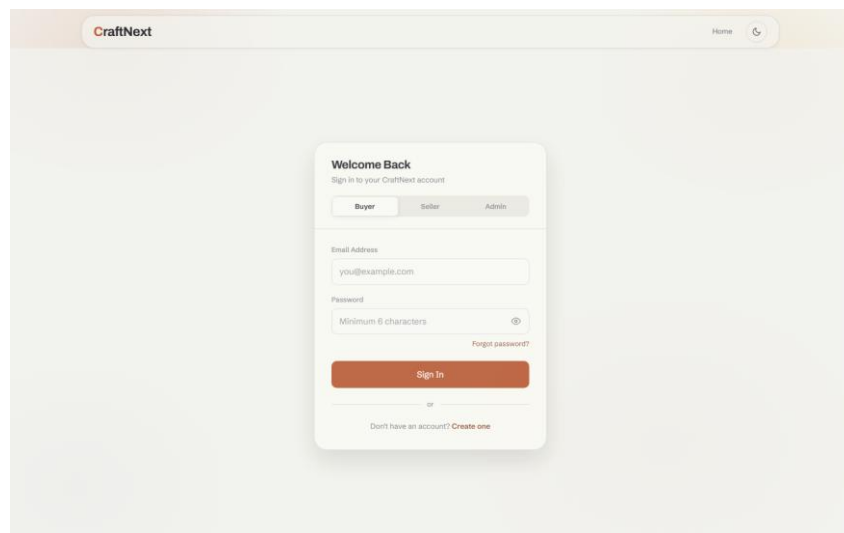


Fig. 4.8 – Login and Registration Interface

4.7.2 User Dashboard

The dashboard provides access to the major system features, including artisan browsing and quick access to ongoing orders.

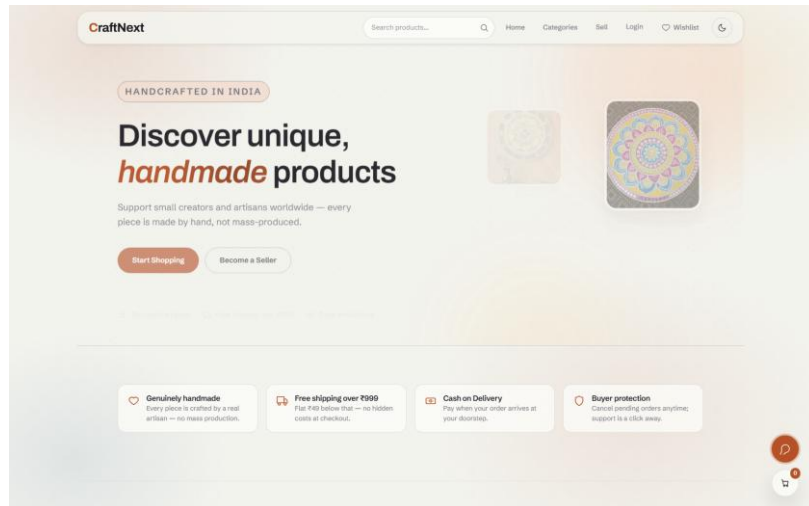


Fig. 4.9 – User Dashboard

4.7.3 Product Details Interface

The product interface displays the artisan details, craft story, price, stock count, and current ordering status.

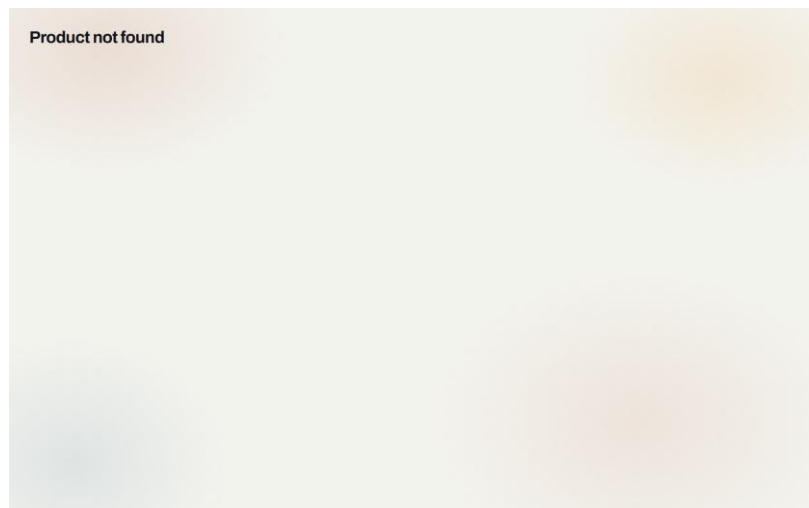


Fig. 4.10 – Product Details Interface

4.7.4 Artisan Dashboard

The artisan can monitor incoming customer requests and stock entries, and manage order status through this dashboard.

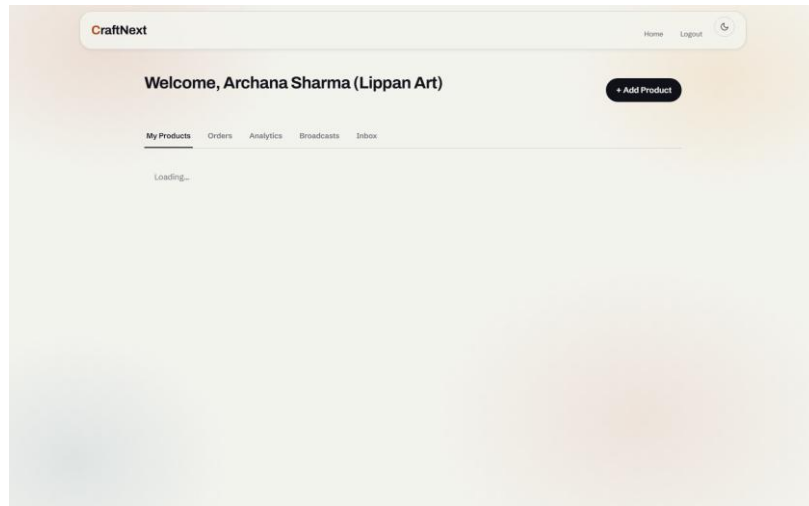


Fig. 4.11 – Artisan Dashboard

4.7.5 Order Management Interface

Shows pending, in-progress and completed orders, allowing the user to track order progress in real time.

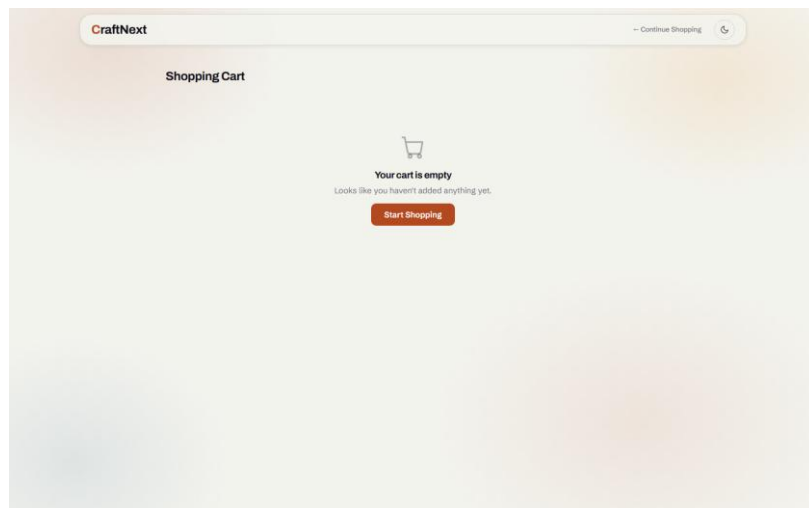


Fig. 4.12 – Order Management Interface

CHAPTER 5

RESULTS AND TESTING

5.1 Functional Results

The implemented system provides the following functionality:

- Authentication – Users can authenticate through the login system.
- Artisan Discovery – Users can view available artisans.
- Artisan Filtering – Users can filter artisans by category.
- Cart Storage – Users can add handcrafted items to cart.
- Token Generation – The system generates a unique order token.
- Order Tracking – Users can monitor their order position.
- Artisan Management – Artisans can view and manage customer requests.
- Order Management – Artisans can update order status.
- Real-Time Communication – Users receive product and order updates.

5.2 Testing Methodology

Testing is performed to verify that the application behaves according to the specified requirements. The following types of testing are included.

5.2.1 Functional Testing

Checks whether each function produces the expected result.

5.2.2 Integration Testing

Checks communication between the frontend, backend, database, authentication service and real-time service.

5.2.3 UI Testing

Checks buttons, forms, navigation, layout and responsiveness.

5.2.4 Authentication Testing

Checks valid and invalid login scenarios.

5.2.5 Concurrency Testing

Checks stock validation and concurrent order progression.

5.2.6 Real-Time Testing

Checks whether order updates are communicated correctly.

5.3 Test Cases

Table 5.1 – General Functional Testing

Test ID	Test Case	Expected Result
T01	Open application	Application loads
T02	Register user	User registration succeeds
T03	Login with valid credentials	User enters dashboard
T04	Login with invalid credentials	Error displayed
T05	Browse artisans	Artisans displayed
T06	Search artisan	Matching artisans displayed
T07	Filter artisan	Matching category displayed
T08	Select artisan	Artisan details displayed
T09	Add to cart	Cart entry created
T10	Generate order token	Unique token displayed
T11	View order status	Current status displayed
T12	Update order status	Status changes in real time
T13	Place order	Order created and stored
T14	Artisan views order	Order displayed on dashboard
T15	Artisan updates order	Status updated in database
T16	User tracks order	Updated status displayed to user

5.4 Authentication Testing

Table 5.2 – Authentication Test Cases

Test ID	Scenario	Expected Result	Status
A01	Valid login	Login successful	Pass
A02	Invalid password	Error displayed	Pass
A03	Empty email	Validation error	Pass
A04	Empty password	Validation error	Pass
A05	Logout	Session terminated	Pass

5.5 Queue & Stock Testing

Table 5.3 – Stock and Queue Management Test Cases

Test ID	Scenario	Expected Result
Q01	Order available craft	Entry created
Q02	Order unavailable craft	Appropriate message
Q03	Generate order token	Unique token
Q04	Multiple users order	Correct sequence
Q05	Artisan serves first order	Next order moves forward
Q06	User checks order status	Correct position displayed
Q07	Order status changes	User receives update

5.6 Order Testing

Table 5.4 – Order Management Test Cases

Test ID	Scenario	Expected Result
O01	Create order	Order stored
O02	Artisan views order	Order visible
O03	Change Pending → In Progress	Status updated
O04	Change In Progress → Completed	Order completed
O05	User checks order	Latest status displayed

5.7 Real-Time Testing

Real-time functionality is especially important because the project aims to provide live stock and order information.

Test Scenario: User A orders a handcrafted terracotta vase with Token A101 at Current Position 5. The artisan processes one order, the system updates the queue, and User A receives an updated position of 4. The user interface reflects the new status without requiring a manual page refresh.

5.8 User Interface Testing

The GUI is tested on desktop, laptop, tablet and mobile browsers, and is designed to be responsive across these devices [12]. The following aspects are checked:

- Text readability.
- Button visibility.
- Navigation.
- Form usability.
- Responsive layout.
- Image loading.

- Error messages.
- Order timeline display.

5.9 Overall Results

The CraftNext system provides an integrated approach to digital craft commerce and artisan management. The major outcomes include:

- Digital artisan marketplace participation.
- Unique order tokens and dynamic tracking.
- Live stock-position tracking and reservation.
- Verified artisan discovery and provenance.
- Digital order and inventory management.
- Artisan-side fulfillment control dashboard.
- Real-time status and delivery communication.
- Responsive mobile and desktop interface.
- Reduced dependency on physical middlemen.
- Better organization of customer requests.

These results are consistent with reduced intermediary costs, improved inventory management, enhanced convenience, better service efficiency and organized order processing as the major benefits of the system.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The CraftNext – Digital Marketplace for Handcrafted Products was developed to address the common problem of inefficient physical supply chains and middleman exploitation in artisan handicraft environments. Traditional craft systems require artisans to depend on multiple tiers of brokers and wholesale dealers, creating severe margin loss, delayed payments and economic precarity. The proposed CraftNext platform introduces a digital alternative by allowing users to discover artisans, select handcrafted products, execute orders digitally and monitor fulfillment progress.

One of the major contributions of the project is the integration of artisan discovery and order management. Instead of treating craft selling as an isolated process, CraftNext allows users to first discover an appropriate artisan and then interact with that artisan through the digital platform.

The system also introduces digital token generation and atomic stock management. When a user places an order, the system assigns an order token that represents the transaction, providing greater transparency compared with manual paper receipts. Real-time communication further improves the system because users can receive updates regarding order and delivery status, reducing the need to continuously inquire about progress.

The artisan interface provides creators with tools for managing customer requests, product stock counts and order status, so the system attempts to improve both sides of the craft commerce interaction.

The project was developed using HTML5, Vanilla CSS, and JavaScript for frontend development, Node.js and Express.js for backend services, MongoDB for database management, Razorpay for payment processing, and Google Gemini AI for conversational shopping assistance. The project also provided practical experience in requirement analysis, system design, frontend development, backend development, database management, authentication, real-time communication and testing.

Overall, CraftNext demonstrates that a digital marketplace approach can provide a more convenient and organized craft commerce experience compared with traditional middleman chains, resulting in fair artisan earnings, reduced intermediary costs, improved service efficiency and enhanced customer satisfaction.

6.2 Future Scope

6.2.1 Mobile Application

A dedicated Android/iOS application can provide easier access, allowing users to browse crafts, track orders, receive push notifications, view artisans and manage orders.

6.2.2 Online Payment Escrow

Payment gateways can be integrated to support escrow smart payouts, UPI, credit/debit cards, digital wallets and net banking, releasing payments to artisans upon customer delivery.

6.2.3 Push Notifications

Notifications and SMS alerts can inform users when their order is accepted, being processed, dispatched or delivered.

6.2.5 AI-Based Artisan Recommendation

The system could recommend artisans and crafts according to user preferences, previous selections, rating, availability, craft materials and category.

6.2.6 Artisan Analytics

Artisans could receive dashboards showing the number of customers, peak seasonal demand, average fulfillment time, completed orders, cancelled orders and average sales volume, supporting decisions on production trends and craft performance.

6.2.7 Multi-Artisan Cluster Support

The system can be expanded to support large craft clusters such as state cooperatives, handloom societies, pottery villages and tribal development federations.

6.2.8 Multi-Location & Multilingual Support

A single user account could access artisans across multiple regions — for example, Gujarat Kutch crafts with Artisan A as the Lippan Studio and Artisan B as the Terracotta Potter, with regional language voice assistance.

6.3 Limitations

- Internet connectivity is required.
- The current system primarily operates through a web interface.
- Artisan information must be kept updated.
- Real-world large-scale deployment may require additional infrastructure.
- Advanced AI-based 3D room preview is not part of the current core implementation.
- Logistics courier tracking may require additional third-party integration.
- User adoption depends on awareness and ease of use.

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WEEKLY REPORT

Week 1 – Project Selection

- Identified the problem of middleman exploitation in handicraft supply chains.
- Discussed possible project ideas.
- Selected digital craft marketplace and artisan empowerment as the project domain.
- Defined the initial project concept.

Week 2 – Requirement Analysis

- Studied existing artisan commerce and e-commerce systems.
- Identified problems with traditional craft distribution channels.
- Defined user requirements.
- Defined artisan requirements.
- Prepared initial functional requirements.

Week 3 – System Design

- Designed application workflow.
- Identified project modules.
- Planned database structure.
- Prepared initial DFD and ER diagram.
- Designed UI structure.

Week 4 – Frontend Development

- Created web application layouts.
- Developed common UI components.
- Designed dashboard.
- Implemented artisan and product cards.
- Added category filtering.

Week 5 – Authentication

- Integrated authentication.
- Implemented registration/login.
- Tested user sessions.
- Added access control.

Week 6 – Backend Development

- Created Node.js backend.
- Configured Express.js.
- Developed REST APIs.
- Connected frontend and backend.

Week 7 – Database Integration

- Configured database systems.
- Designed tables and collections.
- Implemented database operations.
- Connected application APIs with database.

Week 8 – Stock and Queue Management

- Implemented digital stock management.
- Developed order token-generation mechanism.
- Added order position tracking.
- Implemented artisan order management.

Week 9 – Real-Time Functionality

- Integrated real-time communication.
- Implemented order updates.
- Tested status synchronization.

Week 10 – Order Management

- Implemented order creation.
- Added artisan order management.
- Added order status updates.
- Tested customer/artisan interaction.

Week 11 – Testing

- Performed functional testing.
- Tested authentication.
- Tested concurrency operations.
- Tested artisan functionality.
- Fixed application errors.

Week 12 – Documentation

- Prepared project report.
- Added diagrams.
- Added screenshots.
- Prepared testing documentation.
- Prepared final presentation.